

Assignment 9 Solutions

1.35

$$\begin{bmatrix} -1 & -2 & -1 & -4 \\ 1 & -1 & 2 & 3 \\ -2 & -4 & 1 & -7 \end{bmatrix} \sim \begin{bmatrix} -1 & -2 & -1 & -4 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -3 & -3 \end{bmatrix} \sim \begin{bmatrix} -1 & -2 & -1 & -4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

No solution

2.11

$$\begin{bmatrix} 1 & -2 & -8 & 16 \\ -2 & 5 & 21 & -38 \\ 0 & 2 & 10 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -8 & 16 \\ 0 & 1 & 5 & -6 \\ 0 & 2 & 10 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -8 & 16 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 4 - 2x_3 \\ x_2 &= -6 - 5x_3 \\ x_3 &\text{ is free} \end{aligned}$$

2.18

$$\begin{bmatrix} 1 & 3 & b_1 \\ 2 & 7 & b_2 \\ 1 & 5 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & 2 & b_3 - b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & 0 & b_3 - b_1 - 2(b_2 - 2b_1) \end{bmatrix}$$

$$\begin{aligned} b_3 - b_1 - 2(b_2 - 2b_1) &= b_3 - b_1 - 2b_2 + 4b_1 \\ &= 3b_1 - 2b_2 + b_3 \end{aligned}$$

$$3x - 2y + z = 0$$

3.13

$$\begin{bmatrix} 1 & -5 & 4 \\ -1 & 6 & -3 \\ -2 & 13 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

Yes. 1 pivot per row

4.3

$$\begin{bmatrix} 1 & -3 & -6 & -1 \\ -2 & 7 & 15 & 3 \\ 1 & -2 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -6 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -6 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -6 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x_1 = -3x_3$$

$$x_2 = -3x_3$$

x_3 is free

$$x_4 = 0$$

$$3\vec{v}_1 + 3\vec{v}_2 - \vec{v}_3 = \vec{0}$$

5.3

$$\begin{bmatrix} 5 & 7 & 10 \\ 2 & -1 & 4 \\ 0 & -3 & 0 \end{bmatrix}$$

$$\text{dom}(T) = \mathbb{R}^3$$

$$\text{cod}(T) = \mathbb{R}^3$$

5.13

$$\begin{bmatrix} 1 & -1 & 2 & 6 \\ -2 & -1 & 3 & 12 \\ -2 & 2 & -3 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 6 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$$

unique

5.19

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) - T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ -3 \\ 4 \end{bmatrix} - \begin{bmatrix} 8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \\ 2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) - T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} -5 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 3 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \frac{1}{2} \left(T\left(\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}\right) - T\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) \right) = \frac{1}{2} \left(\begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 3 \\ -3 \\ 4 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} -4 \\ 2 \\ -5 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -5/2 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 13 & -2 \\ -3 & 3 & 1 \\ 2 & 0 & -5/2 \end{bmatrix}$$

5.21

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 1 & 3 \\ 1 & 5 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

onto, not 1-1

5.26

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

1-1, not onto

5.30 Both (already in echelon form, 1 pivot per col/row)

$$6.4 \begin{bmatrix} 6 & 1 \\ -1 & -3 \\ 8 & 1 \end{bmatrix} \begin{bmatrix} -4 & -8 & -4 \\ 0 & 5 & -1 \end{bmatrix} = \begin{bmatrix} -24 & -43 & -25 \\ 4 & -7 & 7 \\ -32 & -59 & -33 \end{bmatrix} \begin{bmatrix} -4 & 0 \\ -8 & 5 \\ -4 & -1 \end{bmatrix} \begin{bmatrix} 6 & -1 & 8 \\ 1 & -3 & 1 \end{bmatrix} = \begin{bmatrix} -24 & 4 & -32 \\ -43 & -7 & -59 \\ -25 & 7 & -33 \end{bmatrix}$$

$$\begin{bmatrix} -24 & -43 & -25 \\ 4 & -7 & 7 \\ -32 & -59 & -33 \end{bmatrix} + \begin{bmatrix} -24 & 4 & -32 \\ -43 & -7 & -59 \\ -25 & 7 & -33 \end{bmatrix} = \begin{bmatrix} -48 & -39 & -57 \\ -39 & -14 & -52 \\ -57 & -52 & -66 \end{bmatrix}$$

$$6.21 \begin{bmatrix} 2 & -1 \\ 3 & 3 \end{bmatrix}^{-1} = \frac{1}{6 - (-3)} \begin{bmatrix} 3 & 1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/9 \\ -1/3 & 2/9 \end{bmatrix}$$

$$6.25 \begin{bmatrix} 1 & -1 & 1 & 1 & 0 & 0 \\ -3 & 4 & -4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 3 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 & 0 \\ 3 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 4 & 1 & 0 \\ 0 & 1 & 0 & 3 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$6.31 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ -3 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ -3 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ -13 & -5 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -13 & -5 & 1 \end{bmatrix}$$

$$R_2 \leftarrow R_2 + 2R_1$$

$$R_3 \leftarrow R_3 - 2R_1$$

$$R_3 \leftarrow R_3 + 2R_2$$

$$R_4 \leftarrow R_4 + 3R_1$$

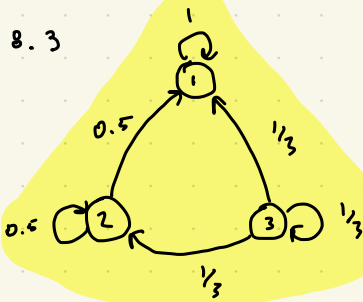
$$R_4 \leftarrow R_4 - 2R_2$$

$$R_4 \leftarrow R_4 + 2R_3$$

$$7.4 \quad \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ -2 & 3 & -13 & -1 & 8 \\ 2 & -4 & 18 & 3 & -2 \\ -3 & 5 & -22 & -4 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & -2 & 10 & 3 & 8 \\ 0 & 2 & -10 & -4 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -2 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 2 & -2 & 1 & 0 \\ -3 & 2 & -2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 4 & 0 & -5 \\ 0 & 1 & -5 & -1 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

8.3



Not regular. $A\vec{e}_1 = \vec{e}_1 \Rightarrow A^k\vec{e}_1 \Rightarrow \vec{e}_1$

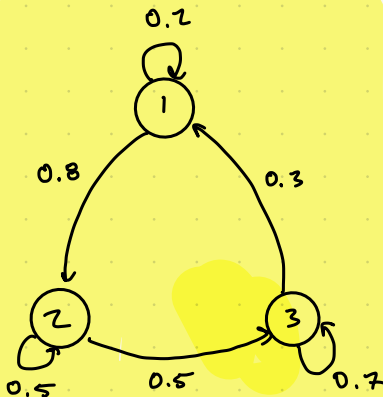
$$A - I = \begin{bmatrix} 0 & 1/2 & 1/3 \\ 0 & -1/2 & 1/3 \\ 0 & 0 & -2/3 \end{bmatrix} \sim \begin{bmatrix} 0 & 1/2 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free
 $x_2 = 0$
 $x_3 = 0$

$$\vec{s} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

unique

8.7



$$A^2 = \begin{bmatrix} 0.04 & 0.15 & 0.27 \\ 0.56 & 0.25 & 0.24 \\ 0.4 & 0.6 & 0.49 \end{bmatrix}$$

$k=2$

$$A - I =$$

$$\begin{bmatrix} -0.8 & 0 & 0.3 \\ 0.8 & -0.5 & 0 \\ 0 & 0.5 & -0.3 \end{bmatrix} \sim \begin{bmatrix} -8 & 0 & 3 \\ 8 & -5 & 0 \\ 0 & 5 & -3 \end{bmatrix} \sim \begin{bmatrix} -8 & 0 & 3 \\ 0 & -5 & 3 \\ 0 & 5 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3/8 \\ 0 & 1 & -3/5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = \frac{3}{8} \cdot \frac{40}{79} = \frac{15}{79}$$

$$x_2 = \frac{3}{5} \cdot \frac{40}{79} = \frac{24}{79}$$

$$x_3 = \frac{40}{79}$$

unique

$$x_1 = \frac{3}{8} x_3$$

$$x_2 = \frac{3}{5} x_3$$

x_3 is free

$$\left(\frac{3}{8} + \frac{3}{5} + 1 \right) x_3 = \left(\frac{15}{40} + \frac{24}{40} + \frac{40}{40} \right) x_1 = \frac{79}{40} x_1 = 1$$

9.7

$$\begin{bmatrix} 1 & -5 \\ -2 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & -5 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = 5x_2 \\ x_2 \text{ is free} \end{array}$$

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$$

$$\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} 5 \\ 1 \end{bmatrix} \right\}$$

$$\begin{array}{l} 9.9 \quad \begin{bmatrix} 0 & -2 & 2 \\ -2 & 3 & -9 \\ -1 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -9 \\ 0 & -2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 7 & -7 \\ 0 & -2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -3x_3 \\ x_2 = x_3 \\ x_3 \text{ is free} \end{array} \end{array}$$

$$\text{Col } A = \text{span} \left\{ \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix} \right\}$$

$$\text{Nul } A = \text{span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$\begin{array}{l} 9.18 \quad \begin{bmatrix} 1 & 3 & -17 \\ -5 & -1 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -17 \\ 0 & 14 & -70 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -17 \\ 0 & 1 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -5 \end{bmatrix} \quad \begin{bmatrix} -17 \\ 15 \end{bmatrix}_{\beta} = \begin{bmatrix} -2 \\ -5 \end{bmatrix} \end{array}$$

$$10.1 \quad \begin{bmatrix} 6 & 1 \\ -3 & 2 \end{bmatrix} * \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \end{bmatrix} \quad \text{Yes, } \lambda = 5$$

$$10.3 \quad A - \lambda I = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{No}$$

$$\begin{array}{l} 10.8 \quad \det(A - \lambda I) = \det \begin{bmatrix} -3-\lambda & 2 \\ -10 & 6-\lambda \end{bmatrix} = (-3-\lambda)(6-\lambda) - (-20) \\ = \lambda^2 + 3\lambda - 6\lambda - 18 + 20 \end{array}$$

$$\begin{array}{l} A - I = \begin{bmatrix} -4 & 2 \\ -10 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{bmatrix} \\ = \lambda^2 - 3\lambda + 2 \quad \lambda = 1, 2 \end{array}$$

$$\text{Nul}(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$A - 2I = \begin{bmatrix} -5 & 2 \\ -10 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{2}{5} \\ 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A - 2I) = \text{span} \left\{ \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}$$

10.9 $\lambda = 1, -3, -4$

$$\text{Nul}(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$A + 3I = \begin{bmatrix} 4 & 16 & -12 \\ 0 & 0 & -2 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A + 4I = \begin{bmatrix} 5 & 16 & -12 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A + 3I) = \text{span} \left\{ \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\text{Nul}(A + 4I) = \text{span} \left\{ \begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix} \right\}$$

3.17 True $\vec{b} \in \mathbb{R}^m$ and $\vec{b} \notin \text{Col}(A)$

3.25 True Fewer pivots than columns

4.13 True $\text{rank}(A) = \text{rank}(A^T)$ (alt. dep. relation in A^T becomes row ops in A)

5.40 False $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

6.35 False $A^2 = A^2$

6.40 True $A = AB^{-1}B = AA^{-1}B = B$

6.55 False $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

6.56 True A^T is not invertible because columns are L.D.

6.57 False $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

6.63 True We stated this

7.5 True

$$\det(EA) = \det(E) \det(A) = 1$$

$$\Downarrow$$

$$\det(A) = 1 / \det(E)$$

$$\det(E) = 3 \Rightarrow \det(A) = \frac{1}{3}$$

8.15 True

Steady state is unique

8.23 True

$$D_{ii} = \mathbf{1}^T A \mathbf{e}_i$$

9.29 True

We stated this

9.30 False

Should be "smallest"

10.22 True

We stated this

10.23 False

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = 3 \quad \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

10.24 False

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

1.65

$$(a) \quad h = 2/3 \quad k = 100$$

$$(b) \quad h = 0 \quad k = 0$$

$$(c) \quad h = 2/3 \quad k = 9/2$$

2.26

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \quad \text{defined by } z = 0$$

$$2 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 0 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} \right\} \quad \text{with } z = 0$$

(a)

$$4.24 \begin{bmatrix} 1 & 1 & 0 & -1 \\ -2 & -1 & -1 & -6 \\ -3 & -2 & 0 & 0 \\ -1 & 0 & -1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & -1 & -8 \\ 0 & 1 & 0 & -3 \\ 0 & 1 & -1 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & -1 & -8 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 14 \end{bmatrix}$$

$$\vec{v}_1 \notin \text{span} \{ \vec{v}_2, \vec{v}_3, \vec{v}_4 \}$$

$$(b) [\vec{v}_1 \vec{v}_3 \vec{v}_4] \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

L.I.

(c) L.I., subset of L.I. are
L.I.

$$5.51 \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$6.67 \quad AB^T \times A^{-1}B = I$$

$$B^T \times A^{-1}B = A^{-1}$$

$$\times A^{-1}B = (B^{-1})^T A^{-1}$$

$$\times A^{-1} = (B^{-1})^T A^{-1} B^{-1}$$

$$\times = (B^{-1})^T A^{-1} B^{-1} A$$

$$6.68 \quad A(C^{-1}(AB)^T)^T C =$$

$$A((AB)^T)(C^{-1})^T C =$$

$$\cancel{A} \cancel{A} \cancel{B} \cancel{(C^T)^{-1}} \cancel{C^T} = B$$

$$6.73 \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix} \Rightarrow A = \begin{bmatrix} a & -2 \\ c & 3 \end{bmatrix}$$

$$\det A = 3a + 2c = 1$$

$$A^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow$$

$$a + c = -2$$

$$\begin{bmatrix} 1 & 1 & -2 \\ -3 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 7 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -7 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

$$8.26 \quad (a) T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad (b) T^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad T^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(c) \text{ Not regular} \quad T^{2023} = T^{2022} T = T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$T^k = T^{k \bmod 3} = T \text{ or } T^2 \text{ or } I$$

$$(d) T - I = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = x_3 \\ x_2 = x_3 \\ x_3 \text{ is free} \end{array} \quad \begin{array}{l} 3x_3 = 1 \\ x_3 = 1/3 \end{array} \quad \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

unique, only one free variable

$$9.40 \quad \begin{bmatrix} 1 & -4 & b_1 \\ -4 & 17 & b_2 \\ -1 & -2 & b_3 \end{bmatrix} \xrightarrow{\substack{+4 \\ -16}} \begin{bmatrix} 1 & -4 & b_1 \\ 0 & 1 & b_2 + 4b_1 \\ 0 & 2 & b_3 - b_1 \end{bmatrix} \xrightarrow{-2(b_2 + 4b_1)} \begin{bmatrix} 1 & 4 & k \\ 0 & 1 & k \\ 0 & 0 & b_3 - b_1 - 2(b_2 + 4b_1) \end{bmatrix}$$

$$b_3 - b_1 - 2b_2 - 8b_1 = -9b_1 - 2b_2 + b_3$$

$$7x - 2y + z = 0$$

$$-9 + 8 + 1 = 0$$

$$36 - 34 - 2 = 0$$

$$90 - 84 - 6 = 0$$

$$10.35 \quad \det(B(A B)^{-1}(A B)^T A) =$$

$$\cancel{\det(B)} \cdot \frac{1}{\cancel{\det(A)} \cdot \cancel{\det(B)}} \cdot \det(B^T) \cdot \det(A^T) \cdot \cancel{\det(A)} =$$

$$\det(B) \cdot \det(A) = -7$$

$$10.38 \quad -\lambda^3 + 5\lambda^2 - 6\lambda = -\lambda(\lambda^2 - 5\lambda + 6)$$

$$= -\lambda(\lambda - 2)(\lambda - 3)$$

$$\lambda = 0, 2, 3$$

$$A \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A - 0I) = \text{span}$$

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$A - 2I \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A - 2I) = \text{span}$$

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$A - 3I \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Nul}(A - 3I) = \text{span}$$

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}$$